Table 4.1 : Vitamin related Coenzymes

Coenzyme	Derived from Vitamin	Atom (or) group transferred (function)	Dependent enzyme
Thiamine pyrophosphate (TPP)	Thiamine (B_1)	Aldehyde	Transketolase
Flavin mono nucleotide (FMN)	Riboflavin (B_2)	Hydrogen and electrons	L-amino acid oxidase
Flavin adenine dinucleotide (FAD)	Riboflavin (B_2)	Hydrogen and electrons	D-amino acid oxidase
Nicotinamide adenine dinucleotide (NAD) (or) Diphospho pyridine nucleotide (DPN)	Niacin (B_3)	Hydrogen and electrons	Lactate dehydrogenase
Pyridoxal phosphate (PLP)	Pyridoxine (B_6)	Amino	Alanine transminase
Biotin	Biotin (B_7) or (H)	CO_2	Pyruvate carboxylase
Coenzyme A	Pantothenic acid (B_5)	Acyl	Thiokinase
Tetrahydrofolate	Folic acid	one carbon unit (formyl transferase)	----

4.4 Factors influencing Enzyme activity

The important factors that influence the enzyme catalyzed reaction are; pH, Temperature, Substrate concentration, Enzyme concentration, Activators and inhibitors.

4.4 .1 Effect of pH :

1. Change in hydrogen ion concentration influences the enzyme activity. When the velocity is plotted against pH, a bell shaped curve is obtained.

2. The pH at which the enzymatic reaction has maximum velocity is known as **optimum pH.** Most of the enzymes possess optimum pH between 5 and 9. However there are exceptions like pepsin, alkaline phosphatase etc.

Unit 4.indd 108 21-05-2018 15:10:06